
💡 **Context:**

You're analyzing a twice-differentiable function f , and you're at a **critical point** p , meaning:

$$\nabla f(p) = \vec{0}$$

Now you study the **second derivative**, the **Hessian matrix** $S = \nabla^2 f(p)$, to determine the nature of the critical point (min, max, saddle).

📖 **Spectral Theorem (Real Symmetric Case):**

If S is a **real symmetric matrix**, then:

There exists an orthogonal matrix Q such that $Q^T S Q = D$

where:

- D is a **diagonal matrix** with **real eigenvalues** $\lambda_1, \dots, \lambda_n$
 - Q consists of orthonormal eigenvectors of S , i.e., $Q^T = Q^{-1}$
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🔄 **What's Happening in the Diagram:**

You write:

$$[S]_B = Q^T S Q = D$$

This expresses the matrix S in the **orthonormal eigenbasis** $B = \{\vec{v}_1, \dots, \vec{v}_n\}$. That's what diagonalization is — rewriting the matrix in a coordinate system where it acts by simple **scaling**.

Then:

$$S = Q D Q^T$$



This reconstructs the original matrix in the standard basis. You can check it by multiplying out:

$$Q^T S Q = D \Rightarrow S = Q D Q^T$$

Geometric Interpretation:

- The matrix S **acts diagonally** in the eigenbasis. That is, along each eigenvector direction $\vec{v}_i$, the matrix just scales by λ_i .
- This lets you **classify the critical point** by checking the **signs** of the eigenvalues:
 - All $\lambda_i > 0$: **local minimum**
 - All $\lambda_i < 0$: **local maximum**
 - Mixed signs: **saddle**
 - Any $\lambda_i = 0$: **degenerate**, inconclusive

Summary of the Diagonalization Step:

1. Use **Spectral Theorem**: diagonalize symmetric Hessian matrix S
2. Get orthogonal eigenvectors Q , real eigenvalues in D
3. Write $S = Q D Q^T$
4. Use signs of λ_i to classify the critical point

Would you like a full LaTeX write-up of this explanation?